

RAMIFICATION INVARIANTS OF PURE EXTENSIONS

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ABSTRACT. We study ramification invariants of finite separable extensions of valued fields that are pure on a primitive element. In the immediate unbranched case, we prove that the extended trace ideal is generated by minimal-polynomial derivatives of integral primitive elements and that the Dedekind different is its divisorial closure. For Galois extensions, we express this derivative ideal as a product of subgroup ramification ideals. These formulas extend prime-degree calculations to pure extensions of higher degree. We compute the differential module, traces, differentials, ramification ideals, and character Swan ideals of a pure immediate elementary abelian extension of degree p^2 . A totally ramified example of the same degree gives a comparison with the defectless case.

1. INTRODUCTION

Let $(L/K, v)$ be a finite separable extension of valued fields of degree $n > 1$, and let $\mathcal{O}_K$ and $\mathcal{O}_L$ be the corresponding valuation rings. We study how purity allows descriptions of ramification invariants in prime degree to extend to higher degree. Our starting point is the computation of Kähler differentials by Novacoski and Spivakovsky [11] and the description of ramification ideals arising from the work of Dutta and Novacoski [6]. Both express ramification in terms of linear approximations to a primitive element. We show that these approximations also determine the trace ideal and, through trace duality, the Dedekind different. In the Galois case, the resulting product formula incorporates every conjugate-distance level, extending the single-level formula

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in prime degree. The examples show why retaining the full ideals, including their boundary behavior, matters in extensions with defect.

Over complete discretely valued fields, the different and trace are related to the classical ramification groups. In the totally ramified Galois case, let d be the integral different exponent, let G_i be the lower ramification groups, and put $e = [L : K]$. Then

$$d = \sum_{i \geq 0} (|G_i| - 1), \quad \mathrm{Tr}_{L/K}(\mathcal{O}_L) = \mathfrak{m}_K^{\lfloor d/e \rfloor}$$

by the classical formulas in [1]. In this setting the valuation ring is monogenic over the base valuation ring, and the different is generated by a minimal-polynomial derivative and annihilates $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$.

For more general valuations, defect and nonprincipal ideals require additional care. For a finite unbranched extension, write

$$d_{L/K} = \frac{[L : K]}{e(L/K)f(L/K)}.$$

Here $e(L/K)$ and $f(L/K)$ are the value-group index and residue-field degree, respectively. If $p = \mathrm{char} K v > 0$, then $d_{L/K} = p^s$ for an integer $s \geq 0$, called the defect of $(L/K, v)$. An immediate extension has $e = f = 1$, so a nontrivial immediate extension has nontrivial defect. In this situation $\mathcal{O}_L$ is not a finite $\mathcal{O}_K$ -module. Moreover, a value cut may have no least element, and its infimum need not determine the corresponding ideal.

Several approaches extend ramification theory beyond the classical setting. Kato and Abbes–Saito extend conductor theory to imperfect residue fields [2, 3]. Thatte computes prime-degree ramification and conductor ideals, including defect extensions [12, 13]. Kato–Thatte define an upper filtration indexed by ideals for arbitrary henselian valuation rings [14]. The ramification ideals, differential modules, and trace ideals studied by Kuhlmann and collaborators provide further descriptions in arbitrary rank [9, 5, 8].

To describe the role of purity, fix a primitive element θ of L/K , put $n = [L : K]$, and let g be its monic minimal polynomial. We say that the extension is pure on θ if the truncations defined by linear key polynomials $X - c$, $c \in K$, recover the values of all polynomials of degree less than $[L : K]$. For such a generator, set

$$D = v(\theta - K) = \{v(\theta - c) : c \in K\}.$$

The two descriptions that motivate our results are expressed in terms of D . To state them, suppose that the extension is immediate and unbranched: its value group and residue field agree with those of K ,

and v has a unique extension from K to L . For a final segment S of vL , write

$$I_S = \{0\} \cup \{x \in L^\times : vx \in S\}.$$

Here $\text{Final}(E)$ denotes the smallest final segment containing a set $E \subseteq vL$. Since D is an initial segment in the immediate case, $-D$ is already a final segment. Set

$$\alpha = -D, \quad \beta = \text{Final}(v(g'(\theta)) - nD),$$

where $nD = \{nd : d \in D\}$. The computation of Novacoski and Spivakovsky [11] then gives the description of the module of Kähler differentials

$$(1) \quad \Omega_{\mathcal{O}_L/\mathcal{O}_K} \simeq I_\alpha/I_\beta.$$

For the second description assume, in addition, that L/K is Galois and put $G = \text{Gal}(L/K)$. For a nontrivial subgroup $H \leq G$, its ramification ideal and its least conjugate-distance value are

$$I_H = \left(\frac{\sigma x - x}{x} : x \in L^\times, \sigma \in H \right) \mathcal{O}_L,$$

$$\beta_H = \min_{\sigma \in H \setminus \{1\}} v(\sigma\theta - \theta).$$

The work of Dutta and Novacoski [6] relates purity and distances between conjugates to ramification ideals. The resulting subgroup formula, stated in [10], is

$$(2) \quad I_H = I_{\beta_H - D}.$$

Formulas (1) and (2) show how purity relates these two invariants to the approximation set D . The results below extend this description to the trace ideal and the different.

We first consider the trace ideal and the different. For an $\mathcal{O}_L$ -submodule $J \subseteq L$, write $\mathcal{O}_L : J = \{x \in L : xJ \subseteq \mathcal{O}_L\}$. Set

$$T = \text{Tr}_{L/K}(\mathcal{O}_L)\mathcal{O}_L,$$

$$\mathfrak{D}_{\text{prim}} = (h'_b(b) : b \in \mathcal{O}_L, K(b) = L)\mathcal{O}_L,$$

$$\mathfrak{C} = \{x \in L : \text{Tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K\},$$

$$\mathfrak{D} = \mathcal{O}_L : \mathfrak{C},$$

Here h_b is the monic minimal polynomial of b over K ; $\mathfrak{C}$ is the complementary module and $\mathfrak{D}$ the Dedekind different. We call $\mathfrak{D}_{\text{prim}}$ the primitive-generator different and write $J^v = \mathcal{O}_L : (\mathcal{O}_L : J)$ for the divisorial closure of a fractional ideal J . Our first main result identifies these ideals in the pure immediate case.

Theorem 1.1. *Assume that $(L/K, v)$ is immediate, unibranched, and pure on θ . Then*

$$T = \mathfrak{D}_{\text{prim}}, \quad \mathfrak{C} = \mathcal{O}_L : \mathfrak{D}_{\text{prim}}, \quad \mathfrak{D} = \mathfrak{D}_{\text{prim}}^v.$$

The principal assertion of Theorem 1.1 is $T = \mathfrak{D}_{\text{prim}}$: purity identifies the trace ideal with the ideal generated by minimal-polynomial derivatives of integral primitive elements. The complementary-module and Dedekind-different identities are then consequences of trace duality. Indeed, $\mathfrak{C} = \mathcal{O}_L : T$ and $\mathfrak{D} = T^v$ are already known under the more general hypothesis $vL = vK$ [5, Theorem 5.4]. Thus the theorem supplies an explicit approximation-theoretic description of the ideal entering those identities. The proof, given in Theorem 5.1, uses linear truncations and the interpolation formula for the trace. If g' is constant, it also gives explicit trace generators (Corollary 5.2).

In the Galois case, the factorization of $g'(\theta)$ into conjugate differences leads to a corresponding factorization of the primitive-generator different. This connects the trace calculation to the subgroup ramification ideals of (2).

Theorem 1.2. *Assume that $(L/K, v)$ is immediate, unibranched, Galois, and pure on θ . Then*

$$\mathfrak{D}_{\text{prim}} = \prod_{\sigma \in G \setminus \{1\}} I_{\langle \sigma \rangle}.$$

This is proved in Theorem 4.3. In degree p , all nonidentity conjugate differences have the same value, so the product reduces to I_G^{p-1} . In higher degree, distinct values may occur, and each associated ramification ideal appears with the multiplicity of that value. Corollary 4.5 characterizes when the product still equals $I_G^{[L:K]-1}$.

The example in [15, Section 5.2] provides a concrete application beyond prime degree: it has depth one, degree p^2 , and defect p^2 . Its purity allows the preceding formulas to be applied despite the nontrivial defect. With the notation of Section 7, we obtain

$$T = \mathfrak{D}_{\text{prim}} = t^{p-1} \mathfrak{m}_L, \\ \mathfrak{D} = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} = t^{p-1} \mathcal{O}_L.$$

By Proposition 6.1, the same quotient also describes $\Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log}$. The two subgroup ramification ideals are $\mathfrak{m}_L$ and $t\mathfrak{m}_L$. Put $A = \mathcal{O}_{K^h}$, where K^h is the henselization of K . The degree- p character quotients of LK^h/K^h have Swan ideals $\mathfrak{m}_A$ and $t^p \mathfrak{m}_A$. These conductor computations apply the existing prime-degree theory of Thatte and the ideal-indexed upper filtration of Kato–Thatte [12, 14]; they do not require a higher-degree

parameter theory. The character conductors determine the logarithmic upper filtration. The norm of $t\mathfrak{m}_L$ in the full extension is instead $t^{p^2}\mathfrak{m}_A$, showing that this norm need not be a character conductor. We also compute a totally ramified degree- p^2 example to compare these formulas with the defectless case.

Table 1 summarizes these formulas and their scope. Except for the conductor rows, the three cases concern finite separable immediate unbranched Galois extensions. The degree- p case has $p = \text{char } Kv$; the pure case has degree n and assumes purity on θ ; the general case drops that assumption. Retain the notation D , α , β , and T introduced above, and put

$$b_\sigma = v(\sigma\theta - \theta), \quad b_H = \min_{\sigma \in H \setminus \{1\}} b_\sigma,$$

$$J = \prod_{\sigma \neq 1} I_{b_\sigma - D}.$$

All ideal colons in the table are taken in L . For the conductor rows assume a henselian rank-one base of characteristic p . For an order- p character χ , let $\mathcal{H}_\chi$ be its Swan ideal and put $Q_\chi = L^{\ker \chi}$. If $\mathfrak{a}$ is a nonzero proper ideal in the valuation ring of a separable closure, write

$$\begin{aligned} N_\chi &= (N_{Q_\chi/K} I_{\text{Gal}(Q_\chi/K)}) \mathcal{O}_K, \\ B_\mathfrak{a} &= \{\chi \neq 0 : \mathfrak{a} \cap \mathcal{O}_K \subsetneq \mathcal{H}_\chi\}, \\ K_\mathfrak{a} &= \bigcap_{\chi \in B_\mathfrak{a}} \ker \chi. \end{aligned}$$

Thus $B_\mathfrak{a}$ consists of the nonzero characters that vanish on the upper ramification image at index $\mathfrak{a}$. Intersecting their kernels recovers that image; an empty intersection means G . The strict inclusion retains the behavior at the conductor boundary. In the last row, $U_\mathfrak{a}$ denotes the image in G of the logarithmic upper ramification group indexed by $\mathfrak{a}$. This row assumes G elementary abelian, and $C_\mathfrak{a}$ abbreviates $\mathfrak{a} \cap \mathcal{O}_K \supseteq \mathcal{H}_\chi$ in degree p . A dash indicates that no formula in approximation data is asserted here; it is not a claim that the invariant is unknown in the literature.

The pure differential formula comes from [11], and the prime-degree description is included in [4, 5]. The primitive different and trace formulas are proved in Theorems 4.3 and 5.1. The general trace-duality entries are [5, Theorem 5.4]; the norm and character formulas are Propositions 4.6 and 6.2, using [12, 14] for the conductor identification.

The paper is organized as follows. Section 2 fixes notation and proves the affine-presentation criterion. Sections 3–5 treat differentials, ramification ideals, differentials, norms, and traces. Section 6 recalls the

TABLE 1. Invariant formulas and their scope.

Invariant	Degree p	Pure case	General case
Kähler differentials Ω	I_G/I_G^p	I_α/I_β	—
Annihilator $\text{ann } \Omega$	$\mathcal{O}_L \cap (I_G^p : I_G)$	$\mathcal{O}_L \cap (I_\beta : I_\alpha)$	—
Primitive different $\mathfrak{D}_{\text{prim}}$	I_G^{p-1}	J	—
Trace ideal T	I_G^{p-1}	J	—
Complementary module $\mathfrak{C}$	$\mathcal{O}_L : I_G^{p-1}$	$\mathcal{O}_L : J$	$\mathcal{O}_L : T$
Dedekind different $\mathfrak{D}$	$(I_G^{p-1})^v$	J^v	T^v
Ramification ideal I_H	I_{b_G-D}	I_{b_H-D}	—
Norm value $v N(x)$	$p vx$	$n vx$	$n vx$
Swan ideal $\mathcal{H}_\chi$	N_χ	N_χ	N_χ
Upper image $U_{\mathfrak{a}}$	G if $C_{\mathfrak{a}}$; $\{1\}$ otherwise	$K_{\mathfrak{a}}$	$K_{\mathfrak{a}}$

conductor results used in Section 7, where the two examples are computed.

2. PRELIMINARIES

Write valuations additively, $v0 = \infty$, and put $\mathcal{O}_K = \{x \in K : vx \geq 0\}$, $\mathfrak{m}_K = \{x : vx > 0\}$. The residue field is $Kv = \mathcal{O}_K/\mathfrak{m}_K$. An extension is immediate if $vL = vK$ and $Lv = Kv$, and unbranched if the valuation of K has a unique prolongation to L . Henselian bases give unbranched algebraic extensions. For a value final segment S , write $I_S = \{0\} \cup \{x \in L^\times : vx \in S\}$. Expressions defining S always include final-segment generation. For fractional ideals put

$$\mathcal{O}_L : J = \{x \in L : xJ \subseteq \mathcal{O}_L\}, \quad J^v = \mathcal{O}_L : (\mathcal{O}_L : J).$$

In dense rank one, a boundary in the value group becomes closed under this divisorial operation.

Let $L = K(\theta)$ be separable of degree $n > 1$, with monic minimal polynomial g . Put $D_1 = v(\theta - K)$ and

$$P = \sum_j P_{j,c}(X - c)^j,$$

$$\nu_c(P) = \min_j \{vP_{j,c} + jv(\theta - c)\}.$$

Purity on θ means that every P with $\deg P < n$ admits $c \in K$ such that $\nu_c(P) = vP(\theta)$. If $v(\theta - d) \geq v(\theta - c)$, expanding $X - c = (X - d) + (d - c)$ gives $\nu_d(P) \geq \nu_c(P)$. Thus finitely many polynomial values can be recovered at once. In the immediate case D_1 has no maximum; choose $t_c \in K$ with $vt_c = -v(\theta - c)$ and set $u_c = t_c(\theta - c)$.

Proposition 2.1. *Let $(L/K, v)$ be finite, separable, immediate, and unibranched, and let $L = K(\theta)$. Purity on θ is equivalent to the existence of an exhaustive nested presentation*

$$\mathcal{O}_L = \bigcup_{\alpha} \mathcal{O}_K[b_{\alpha}],$$

$$b_{\alpha} = r_{\alpha}(\theta - c_{\alpha}) \in \mathcal{O}_L, \quad r_{\alpha} \in K^{\times}, \quad c_{\alpha} \in K.$$

For a pure extension, the normalized linear generators satisfy all the presentation hypotheses of [5, Theorem 1.1], including compatible scaling and divisibility. Conversely, such an affine presentation implies purity whenever $vL = vK$, without requiring immediacy.

Proof. Assume first that the extension is pure. For $y = P(\theta) \in \mathcal{O}_L$, $\deg P < n$, choose c with $\nu_c(P) = vy$. If $P = \sum_i P_{i,c}(X - c)^i$, then $v(P_{i,c}t_c^{-i}) \geq vy \geq 0$, so $y = \sum_i (P_{i,c}t_c^{-i})u_c^i \in \mathcal{O}_K[u_c]$. This proves exhaustion. Order a cofinal family by $\delta_c = v(\theta - c)$ and write $u_c = t_c(\theta - c)$. If $\delta_c \leq \delta_d$, then

$$u_c = \frac{t_c}{t_d}u_d + t_c(d - c),$$

$$v(t_c/t_d) = \delta_d - \delta_c \geq 0,$$

$$v(t_c(d - c)) \geq 0.$$

Thus both transition coefficients belong to $\mathcal{O}_K$, and $\mathcal{O}_K[u_c] \subseteq \mathcal{O}_K[u_d]$. To meet the theorem's additional requirement that its scaling elements lie in $\mathcal{O}_K$, choose a common $k \in K^{\times}$ with $v(kt_c) \geq 0$ and set $a_c = kt_c$. Such a common bound exists: unibranchedness and separability give $\delta_c \leq v(\theta - \tau\theta)$ for any fixed nonidentity K -embedding τ , and the value group of a finite algebraic extension is cofinal over vK . Choose vk at least this bound. The ratios $a_c/a_d = t_c/t_d$ are unchanged. Together with $\mathcal{O}_L = \bigcup_c \mathcal{O}_K[u_c]$, this verifies all the presentation requirements of that theorem.

Conversely, suppose $vL = vK$ and the affine presentation exists. Take a nonzero polynomial $f \in K[X]$ of degree less than n , choose $a \in K^\times$ with $va = v(f(\theta))$, and place $f(\theta)/a$ in some $\mathcal{O}_K[b_\alpha]$. Since b_α is integral and has degree n , its unique expansion of degree less than n has integral coefficients. Substitution gives $\nu_{c_\alpha}(f) \geq va$; the opposite inequality always holds, so equality follows. Finite collections are handled in a common larger algebra.

The compatible affine maps in [5, Theorem 1.1] put every generator on a cofinal tail in the form $r_\alpha(\theta - c_\alpha)$ after choosing one of them as θ . Consequently, in this equal-value-group setting, existence of such a presentation expresses purity on a suitable primitive generator. It does not assert purity on every primitive generator, nor does this argument remove the hypothesis $vL = vK$ for the other pure classes. $\square$

Indeed, [5, Theorem 1.1] gives $\Omega_{\mathcal{O}_L/\mathcal{O}_K} \simeq U/UV$, where

$$\begin{aligned} U &= (kt_c : c \in K)\mathcal{O}_L, \\ V &= (h'_{u_c}(u_c) : c \in K)\mathcal{O}_L. \end{aligned}$$

Here h_{u_c} is the minimal polynomial of u_c . Writing $I_\alpha = (t_c : c \in K)\mathcal{O}_L$ and cancelling the common factor k yields

$$\Omega_{\mathcal{O}_L/\mathcal{O}_K} \simeq I_\alpha/(I_\alpha V).$$

In the pure immediate case the derivative calculation below gives $V = \mathfrak{D}_{\text{prim}}$, and $v(I_\alpha V)$ is generated by $v(g'(\theta)) - nD_1(\theta, K)$. Thus $I_\alpha V = I_\beta$, recovering the corresponding Novacoski–Spivakovsky quotient. The proposition supplies the presentation needed for this alternative derivation; it does not assert that all cases treated in [11] follow from the present immediate argument.

Expressions such as $-D_1$ and $a - D_1$ denote the final segments generated by the negatives or translated negatives of the approximation values. Likewise, mD_1 denotes the set of scalar multiples md , $d \in D_1$, with final-segment generation understood when defining an ideal. This convention also applies when $vL \neq vK$.

The relative differential module $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$ is generated by dx , subject to additivity, the Leibniz rule, and $da = 0$ for $a \in \mathcal{O}_K$. We use

$$\begin{aligned} T &= \text{Tr}_{L/K}(\mathcal{O}_L)\mathcal{O}_L, \\ \mathfrak{C} &= \{x \in L : \text{Tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K\}, \\ \mathfrak{D} &= \mathcal{O}_L : \mathfrak{C}, \\ \mathfrak{D}_{\text{prim}} &= (h'_b(b) : b \in \mathcal{O}_L, K(b) = L)\mathcal{O}_L. \end{aligned}$$

Here h_b is the monic minimal polynomial. The all-elements ideal $\mathfrak{D}_{\text{all}} = (h'_b(b) : b \in \mathcal{O}_L \setminus K)\mathcal{O}_L$ is distinct in composite degree; see [7, Section

4.3]. For a valuation-invariant Galois extension put

$$\mathcal{J}_\sigma = ((\sigma x - x)/x : x \in L^\times) \mathcal{O}_L,$$

$$I_H = \sum_{\sigma \in H \setminus \{1\}} \mathcal{J}_\sigma.$$

Norm ideals mean ideals generated by element norms. The logarithmic module $\Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log}$ additionally has symbols $d \log x$ for $x \in L^\times$, with $d \log(xy) = d \log x + d \log y$, $dx = x d \log x$ for $x \in \mathcal{O}_L \setminus \{0\}$, and $d \log a = 0$ for $a \in K^\times$. Absolute logarithmic differentials omit these relative base relations.

3. DIFFERENTIALS AND SUBGROUP RAMIFICATION IDEALS

The computations of Kähler differentials and ramification ideals are closely related, but they use the approximation data in different ways. In degree p this distinction is largely invisible because the nontrivial conjugate differences have a common value. For pure extensions of arbitrary degree, the distinction becomes essential.

3.1. Kähler differentials. Throughout this subsection, let $(L/K, v)$ be finite, separable, immediate, unbranched, and pure on θ . Put

$$D = D_1(\theta, K) = v(\theta - K).$$

The computation of [11] associates with a complete family of linear key polynomials two final segments α and β . In the pure case,

$$\alpha = -D,$$

and, for a complete family $\mathcal{C}$ of linear key polynomials,

$$\beta = \text{Final}_{vL} \{v(g'(\theta)) - \nu_c(g) : c \in \mathcal{C}\}.$$

Here ν_c is the linear truncation defined in the preliminaries. One has

$$I_\beta \subseteq I_\alpha, \quad \Omega_{\mathcal{O}_L/\mathcal{O}_K} \simeq I_\alpha/I_\beta,$$

and consequently

$$\text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} = I_{\{\gamma \in vL : \gamma + \alpha \subseteq \beta\}} \cap \mathcal{O}_L.$$

Thus the module of differentials is represented on the value group by the part of the numerator final segment α that remains before the denominator final segment β begins.

The segment α is therefore entirely approximation-theoretic, while β contains the algebraic relation supplied by g . The equality $\Omega_{\mathcal{O}_L/\mathcal{O}_K} = 0$ is equivalent to $\alpha = \beta$. When the two cuts are different, the distance between them controls the annihilator of the quotient.

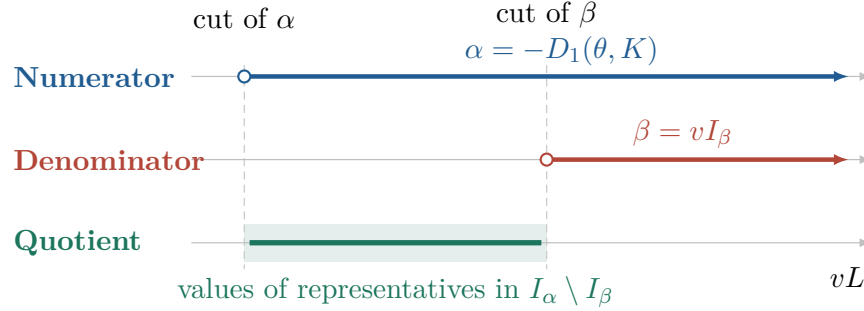


FIGURE 1. The differential quotient I_α/I_β . The blue and red rays lie on separate rows. The shaded interval represents the values of nonzero quotient representatives. The boundary circles are schematic: endpoint membership is determined by the actual final segments.

For Artin–Schreier defect extensions the denominator segment is $-pD$; for Kummer defect generators satisfying $v\theta = 0$ it is $vp - pD$. In the latter case we assume characteristic zero, residue characteristic p , and a primitive p th root of unity in K . Here $D = v(\theta - K)$; Figure 2 illustrates the two translations.

3.2. Ramification ideals. For a pure generator set

$$\beta_H = \min_{\sigma \in H \setminus \{1\}} v(\sigma\theta - \theta).$$

The following formula connects conjugate distances to relative changes of arbitrary field elements.

Theorem 3.1. *Let $(L/K, v)$ be a finite Galois extension, pure on θ , whose valuation is invariant under $G = \text{Gal}(L/K)$ (as is automatic over a henselian base). For every nontrivial $H \leq G$,*

$$(3) \quad I_H = I_{\beta_H - D_1(\theta, K)}.$$

The henselian formulation is stated in [10] as a consequence of [6]. The following argument explains why invariance of the chosen valuation suffices.

Proof. Fix $\sigma \neq 1$ and write $\beta = v(\sigma\theta - \theta)$. For $x = P(\theta) \neq 0$, $\deg P < [L : K]$, choose c with $\nu_c(P) = vx$, and put $\delta = v(\theta - c)$. Valuation invariance gives $v(\sigma\theta - c) = \delta$. Expanding P at c and factoring the difference of two powers gives

$$v(\sigma x - x) \geq vx + \beta - \delta.$$

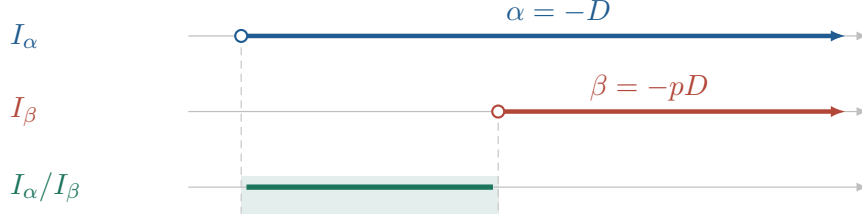
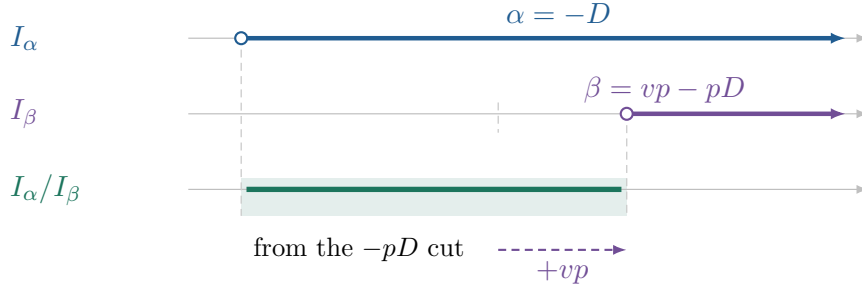
Artin–Schreier defect**Kummer defect**

FIGURE 2. Differential segments in degree p , with $D = D_1(\theta, K)$. Both numerator cuts equal $-D$. The Artin–Schreier denominator is $-pD$; the Kummer denominator has the additional translation by vp . The green bands represent the quotients. The cut positions and open circles are schematic, including the possibility of coincident cuts.

Thus every generator $\sigma x/x - 1$ belongs to $I_{\beta - D_1(\theta, K)}$. Conversely, taking $x = \theta - c$ realizes the value $\beta - v(\theta - c)$ for every $c \in K$. This proves the formula for the ideal generated by one automorphism. Summing over $\sigma \in H \setminus \{1\}$ selects the smallest β_σ , proving (3). $\square$

In a Galois defect extension of degree p , all nontrivial conjugate differences have the same value

$$\delta = v(\theta - \sigma(\theta)).$$

There is only one nontrivial subgroup and hence one ramification ideal, whose value segment is obtained by translating $-D$ by δ .

For a pure Galois extension of degree n , put

$$\beta_\sigma = v(\sigma\theta - \theta) \quad (\sigma \neq 1).$$

Purity does not force these values to coincide. Every distinct level that is realized as a subgroup minimum translates the same final segment $-D_1(\theta, K)$ and produces a ramification ideal.

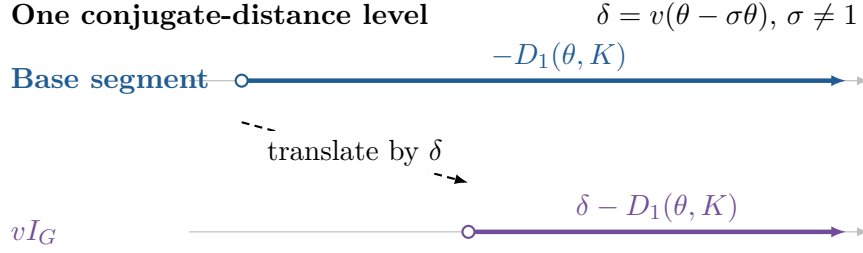


FIGURE 3. Prime-degree ramification translates the approximation segment by the common conjugate-distance value δ . Open boundary markers are schematic.

The differential quotient and the subgroup ramification ideals depend on different operations on the same approximation segment. The former uses the derivative and truncation segment β , while the latter uses translations by subgroup minima β_H . Purity does not imply that the conjugate-distance profile has a single level.

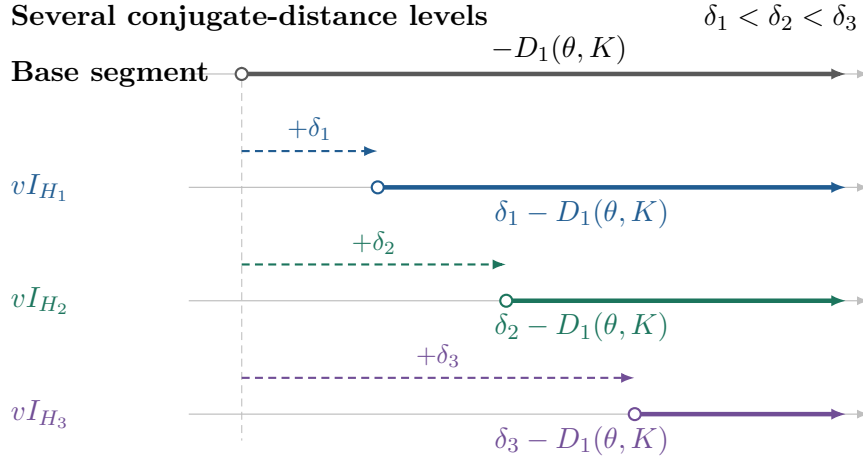


FIGURE 4. Each subgroup minimum δ_i translates the same base segment into vI_{H_i} . The positions are schematic, the value group need not be archimedean, and the boundary markers do not prescribe endpoint membership.

4. THE PRIMITIVE-GENERATOR DIFFERENT AND NORM IDEALS

4.1. The primitive-generator different. The normalized linear presentation of the valuation ring gives a derivative-generator formula for $\mathfrak{D}_{\text{prim}}$.

Theorem 4.1. *Assume that $(L/K, v)$ is finite, separable, immediate, unbranched, and pure on θ . Then*

$$(4) \quad \mathfrak{D}_{\text{prim}}(\mathcal{O}_L/\mathcal{O}_K) = \left((\theta - c)^{1-n} g'(\theta) : c \in K \right) \mathcal{O}_L.$$

Equivalently, its value final segment is the final segment generated by

$$(5) \quad v(g'(\theta)) - (n-1)D_1(\theta, K).$$

Proof. For $c \in K$, choose $t_c \in K$ with $vt_c = -v(\theta - c)$ and put $\theta_c = t_c(\theta - c)$. Purity gives $\mathcal{O}_L = \bigcup_c \mathcal{O}_K[\theta_c]$. For any integral primitive generator b , choose c such that $b = F(\theta_c)$ with $F \in \mathcal{O}_K[X]$. For every nonidentity K -embedding τ of L into a valued algebraic closure, $F(\theta_c) - F(\tau\theta_c)$ is divisible by $\theta_c - \tau\theta_c$ with integral quotient. Here unbranchedness ensures that $\tau\theta_c$ is integral. Since both generators are primitive, multiplying over the same $n-1$ embeddings gives $vh'_b(b) \geq vh'_{\theta_c}(\theta_c)$. Thus $\mathfrak{D}_{\text{prim}}$ is generated by the elements $h'_{\theta_c}(\theta_c)$. The transformed minimal polynomial $h_{\theta_c}(X) = t_c^n g(c + X/t_c)$ gives

$$h'_{\theta_c}(\theta_c) = t_c^{n-1} g'(\theta),$$

whose value is $v(g'(\theta)) - (n-1)v(\theta - c)$. $\square$

The normalized-generator calculation appears in [7, Lemma 4.18]; the proof above establishes it with our primitive-generator convention in arbitrary finite degree. In particular, [7, Proposition 4.19] is the specialization of Theorem 4.1 obtained by replacing the full derivative value by the single conjugate-difference value available in degree p .

Remark 4.2 (Limits of a scaled-generator presentation). Purity in the purely ramified case does not alone imply that $\mathcal{O}_L$ is generated by all integral scalar multiples of a fixed primitive element. For example, take $K = k((t))$, with $\text{char } k \neq 3$, and $\theta^3 = t^2$. The values of $1, \theta, \theta^2$ are distinct modulo vK , so θ is pure. Nevertheless, $vh < v\theta = 2/3$ implies $vh \leq 0$, and

$$\mathcal{O}_K[\theta/h : vh < v\theta] = \mathcal{O}_K[\theta] \neq \mathcal{O}_L : \quad \theta^2/t \in \mathcal{O}_L \setminus \mathcal{O}_K[\theta].$$

Thus the derivative formula above is asserted for the pure immediate case. In a monogenic integral extension $\mathcal{O}_L = \mathcal{O}_K[b]$, the usual formula $\mathfrak{D}_{\text{prim}} = \mathfrak{D} = (h'_b(b))$ applies separately.

4.2. Conjugate-distance factorization. For $\sigma \in G \setminus \{1\}$ put

$$\beta_\sigma = v(\sigma\theta - \theta).$$

Separability gives the exact identity

$$(6) \quad v(g'(\theta)) = \sum_{\sigma \in G \setminus \{1\}} \beta_\sigma.$$

The ramification ideal I_G reflects only

$$\beta_G = \min_{\sigma \neq 1} \beta_\sigma.$$

Combining (5), (3), and (6) gives the product formula below.

For every $\sigma \in G \setminus \{1\}$ define the *conjugate-distance ideal*

$$J_\sigma(\theta/K) := I_{\beta_\sigma - D_1(\theta, K)}.$$

In fact this ideal is intrinsic:

$$J_\sigma(\theta/K) = I_{\langle \sigma \rangle} = \mathcal{J}_\sigma.$$

Indeed, telescoping and valuation invariance give $v(\sigma^i \theta - \theta) \geq v(\sigma \theta - \theta)$, so the subgroup minimum is β_σ . Similarly, $(\sigma^i x - x)/x$ is a sum of integral multiples of conjugates of $(\sigma x - x)/x$, with the same values, whence $\mathcal{J}_{\sigma^i} \subseteq \mathcal{J}_\sigma$. The pure ramification formula proves the assertion.

Theorem 4.3. *Under the hypotheses of Theorem 4.1, assume additionally that L/K is Galois.*

$$\mathfrak{D}_{\text{prim}}(\mathcal{O}_L/\mathcal{O}_K) = \prod_{\sigma \in G \setminus \{1\}} J_\sigma(\theta/K).$$

Consequently,

$$\mathfrak{D}_{\text{prim}}(\mathcal{O}_L/\mathcal{O}_K) \subseteq I_G^{n-1}.$$

If

$$\varepsilon_G(\theta) = \sum_{\sigma \neq 1} (\beta_\sigma - \beta_G),$$

then equality holds precisely when translation by $\varepsilon_G(\theta)$ fixes the final segment $(n-1)(\beta_G - D_1(\theta, K))$.

Proof. The value segment of a product of fractional ideals of a valuation ring is generated by the Minkowski sum of their value segments. Thus the product on the right has value segment

$$\text{Final} \left(\sum_{\sigma \neq 1} \beta_\sigma - (n-1)D_1(\theta, K) \right).$$

Here the final segment generated by the $(n-1)$ -fold Minkowski sum of $-D_1(\theta, K)$ agrees with that generated by $-(n-1)D_1(\theta, K)$: one inclusion is diagonal, and for the other one uses the largest of finitely many elements of the initial segment $D_1(\theta, K)$. The asserted equality now follows from (6) and Theorem 4.1. Since $\beta_\sigma \geq \beta_G$, one has $J_\sigma \subseteq I_G$ for every $\sigma \neq 1$. The final assertion follows by comparing the two value segments. $\square$

In the product, each conjugate-distance ideal occurs with its multiplicity. The ideal I_G is determined by the minimum distance.

In degree p , valuation invariance and the telescoping identity for $\sigma^i\theta - \theta$ imply that all nontrivial conjugate differences have the same value: applying the inequality in both directions, using an inverse of i modulo p , gives equality. This yields the exponent $p - 1$ in the prime-degree factorization. Purity alone does not impose a common conjugate-distance value in higher degree.

Corollary 4.4. *Under the hypotheses of Theorem 4.3, let $I_1, \dots, I_r$ be the distinct ideals $I_{\langle\sigma\rangle}$, and let m_j count the nonidentity automorphisms yielding I_j . Then*

$$\mathfrak{D}_{\text{prim}} = \prod_{j=1}^r I_j^{m_j}, \quad \sum_{j=1}^r m_j = n - 1.$$

The factors and their multiplicities are independent of the chosen pure generator.

Proof. Group the factors in Theorem 4.3, using $J_\sigma(\theta/K) = I_{\langle\sigma\rangle}$. $\square$

Corollary 4.5. *Under the hypotheses of Theorem 4.3, put $S = v(I_G^{n-1})$ and $H(S) = \{\gamma \in vL : S + \gamma = S\}$. Then $H(S)$ is a convex subgroup of vL , and*

$$\mathfrak{D}_{\text{prim}} = I_G^{n-1} \iff \varepsilon_G(\theta) \in H(S).$$

If vL has rank one, equality holds if and only if all nonidentity conjugate differences have the same value.

Proof. The stabilizer of a final segment is a subgroup. If $0 \leq \eta \leq \gamma$ and $S + \gamma = S$, the inclusions $S + \gamma \subseteq S + \eta \subseteq S$ show convexity. The equality criterion is Theorem 4.3. Here S is nonempty and proper: the approximation set has no maximum, and valuation invariance bounds it above by every β_σ . A rank-one ordered group has no nonzero proper convex subgroup, so $H(S) = \{0\}$. The nonnegative summands defining $\varepsilon_G(\theta)$ sum to zero precisely when each is zero. $\square$

4.3. Norms of valuation ideals. The value-theoretic part of the norm computation does not depend on prime degree or purity.

Proposition 4.6. *Let $(L/K, v)$ be a finite unibranched Galois extension of degree n . For every $x \in L$,*

$$v N_{L/K}(x) = n v x.$$

Consequently, if an $\mathcal{O}_L$ -ideal J is generated by $(x_i)_{i \in A}$, then

$$(N_{L/K} J) \mathcal{O}_K = (N_{L/K} x_i : i \in A) \mathcal{O}_K.$$

If $J = I_S$, the value set of the norm ideal is the final segment of vK generated by

$$\{ns : s \in S\}.$$

In particular, for a pure Galois extension,

$$(7) \quad v\left((N_{L/K} I_H) \mathcal{O}_K\right) = \text{Final}_{vK}\left(n(\beta_H - D_1(\theta, K))\right).$$

Proof. Unbranchedness implies $v(\sigma x) = vx$ for every $\sigma \in \text{Gal}(L/K)$, so the product formula for the norm gives $vN(x) = n vx$. If x belongs to the ideal generated by the x_i , then $vx \geq vx_i$ for some i , because a valuation ring has linearly ordered ideals. Hence $vN(x) \geq vN(x_i)$, and $N(x)$ belongs to the ideal generated by the $N(x_i)$. The reverse inclusion is immediate. Passing to value segments proves the remaining assertions. $\square$

The Artin–Schreier and Kummer formulas in [7, Propositions 4.21 and 4.22] are explicit-generator versions of (7); those polynomial expressions do not persist for a general pure generator, but the final-segment formula does.

5. TRACES AND THE DEDEKIND DIFFERENT

We use normalized generators and trace duality to compare the Dedekind different with the primitive-generator different. Purity implies equality with the trace ideal; taking its divisorial closure gives the Dedekind different. Constant derivatives then yield particularly explicit trace generators.

5.1. Trace duality and divisorial closure. For a fractional $\mathcal{O}_L$ -ideal J , put

$$J^v := \mathcal{O}_L : (\mathcal{O}_L : J).$$

This is the divisorial closure of J . In rank one it has the endpoint interpretation described in the preliminaries.

For an integral primitive element b with minimal polynomial h_b , the trace dual of $\mathcal{O}_K[b]$ is

$$(\mathcal{O}_K[b])^* = h'_b(b)^{-1} \mathcal{O}_K[b].$$

This standard monogenic calculation is also used in [12, Lemma 7.1].

For every $x \in \mathfrak{C}$, the inclusion $\mathcal{O}_K[b] \subseteq \mathcal{O}_L$ places x in $(\mathcal{O}_K[b])^*$, hence $h'_b(b)x \in \mathcal{O}_K[b] \subseteq \mathcal{O}_L$. Thus $h'_b(b) \in \mathfrak{D}(\mathcal{O}_L/\mathcal{O}_K)$. Since an inverse fractional ideal is divisorial, it follows that

$$(8) \quad \mathfrak{D}_{\text{prim}}(\mathcal{O}_L/\mathcal{O}_K)^v \subseteq \mathfrak{D}(\mathcal{O}_L/\mathcal{O}_K).$$

Theorem 5.1. *Let $(L/K, v)$ be a finite separable extension of degree $n > 1$. Assume that it is immediate, unbranched, and pure on θ . Put $J = \mathfrak{D}_{\text{prim}}(\mathcal{O}_L/\mathcal{O}_K)$. Then*

$$\begin{aligned} T(\mathcal{O}_L/\mathcal{O}_K) &= J, \\ \mathfrak{C}(\mathcal{O}_L/\mathcal{O}_K) &= \mathcal{O}_L :_L J, \\ \mathfrak{D}(\mathcal{O}_L/\mathcal{O}_K) &= J^v. \end{aligned}$$

These assertions do not require a constant derivative or a rank-one valuation.

Proof. We first prove a trace estimate. For $z = P(\theta) \neq 0$, with $\deg P < n$, choose $c \in K$ such that the linear truncation recovers both P and g' :

$$\nu_c(P) = vz, \quad \nu_c(g') = v(g'(\theta)).$$

Such a common c exists by purity and the monotonicity of linear truncations with the approximation value. Put $\delta = v(\theta - c)$ and $Q = X - c$. Unbranchedness implies that every root of g has distance δ from c . Thus, in $g = Q^n + \sum_{j < n} g_{j,c} Q^j$, each coefficient satisfies $vg_{j,c} \geq (n - j)\delta$ and $\nu_c(g) = n\delta$.

Let R be the remainder of Pg' on division by g . Replacing a term bQ^m , $m \geq n$, by $-bQ^{m-n} \sum_{j < n} g_{j,c} Q^j$ never decreases its weighted value. Euclidean division therefore gives

$$\nu_c(R) \geq \nu_c(Pg') = vz + v(g'(\theta)).$$

Lagrange interpolation identifies $\text{Tr}_{L/K}(z)$ with the coefficient of Q^{n-1} in R : at each root θ_i , $R(\theta_i) = P(\theta_i)g'(\theta_i)$, and the leading coefficient of the interpolation polynomial is $\sum_i P(\theta_i)$. Consequently, when the trace is nonzero,

$$(9) \quad v \text{Tr}_{L/K}(z) \geq vz + v(g'(\theta)) - (n - 1)\delta.$$

For $z \in \mathcal{O}_L$, the right side is at least the value of the generator $t_c^{n-1}g'(\theta)$ of J . Hence $T \subseteq J$.

For the reverse inclusion, take any integral primitive element b with monic minimal polynomial $h \in \mathcal{O}_K[X]$. The identity

$$h'(X) = \text{Tr}_{L/K} \left(\frac{h(X)}{X - b} \right)$$

is coefficientwise: the quotient is a polynomial in $\mathcal{O}_K[b][X]$, and summing over embeddings differentiates the product of the linear factors of h . Every coefficient of h' therefore belongs to $\text{Tr}_{L/K}(\mathcal{O}_L)$. Evaluating at $b \in \mathcal{O}_L$ gives $h'(b) \in T$. Thus $J \subseteq T$.

The complementary-module identity can also be checked directly. We already have $\mathfrak{C} \subseteq \mathcal{O}_L : J$ from (8). If $xJ \subseteq \mathcal{O}_L$ and $y \in \mathcal{O}_L$, apply (9) to $z = xy$. Its right side is nonnegative, since $vy \geq 0$ and $xt_c^{n-1}g'(\theta) \in \mathcal{O}_L$. Thus $\text{Tr}(xy) \in \mathcal{O}_K$ and $x \in \mathfrak{C}$. Inverting gives $\mathfrak{D} = J^v$. $\square$

The passage from the trace ideal to the complementary module is also [5, Theorem 5.4]: for finite unibranched extensions with $vL = vK$, that theorem already gives $\mathfrak{C} = \mathcal{O}_L : T$ and $\mathfrak{D} = T^v$. Purity gives $T = \mathfrak{D}_{\text{prim}}$; the identities for $\mathfrak{C}$ and $\mathfrak{D}$ agree with this more general result.

5.2. Pure extensions with constant derivative.

Corollary 5.2. *Under the hypotheses of Theorem 5.1, suppose that the monic minimal polynomial of θ satisfies $g'(X) = a \in K^\times$. For the normalized generators $u_c = t_c(\theta - c)$, one has*

$$\text{Tr}_{L/K}(\mathcal{O}_L) = (at_c^{n-1} : c \in K)\mathcal{O}_K.$$

Proof. For a monic separable polynomial h of degree n and a root b of degree n , Lagrange interpolation gives

$$\text{Tr}_{K(b)/K}(b^i/h'(b)) = \begin{cases} 0, & 0 \leq i < n-1, \\ 1, & i = n-1. \end{cases}$$

Apply this to $h(X) = g(X + c)$ and $b = \theta - c$. Since $h' = a$,

$$\text{Tr}((\theta - c)^i) = 0 \quad (i < n-1), \quad \text{Tr}((\theta - c)^{n-1}) = a.$$

Thus the normalized generators $u_c = t_c(\theta - c)$ satisfy

$$\text{Tr}(\mathcal{O}_L) = (at_c^{n-1} : c \in K)\mathcal{O}_K.$$

$\square$

Corollary 5.3. *In characteristic p , Corollary 5.2 applies whenever a pure immediate unibranched extension is generated by a root of an irreducible polynomial*

$$X^{p^r} + a_{r-1}X^{p^{r-1}} + \cdots + a_1X^p + a_0X - b, \quad a_0 \neq 0.$$

Its degree can be any p^r for which these hypotheses hold.

Proof. The derivative is the nonzero constant a_0 . $\square$

The corollary is conditional on purity and immediacy; an additive presentation by itself implies neither property.

5.3. Differential annihilators. We work under the assumptions of Theorem 5.1 and set

$$C = \{\gamma \in (vL)_{\geq 0} : \gamma + \alpha \subseteq \beta\},$$

$$S_0 = \text{Final}(v(g'(\theta)) - (n-1)D_1(\theta, K)).$$

By Theorem 5.1,

$$\mathfrak{D}(\mathcal{O}_L/\mathcal{O}_K) = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} \iff S_0^v = C,$$

where $S_0^v = v((I_{S_0})^v)$ denotes the divisorial closure of the value segment (an endpoint closure in rank one). This reduces the equality to a comparison of the derivative-distance cut with the key-polynomial cut defining $\text{ann } \Omega$.

6. SWAN IDEALS AND LOGARITHMIC UPPER RAMIFICATION

Purity gives explicit formulas for the invariants considered above, but the passage to Swan conductors requires additional information. A Swan conductor is attached to a character or, more generally, a representation. Purity is a condition on approximation by elements of K ; it neither specifies such a representation nor implies that the extension is Galois or cyclic of degree p . Even for a pure Galois extension, the norm of I_H from the full field L need not be the Swan ideal of a character: the prime-degree norm theorem uses the norm from the character's quotient field. The two norms have different degrees, as the example in Section 7.2.5 demonstrates.

There is nevertheless a general logarithmic consequence in the immediate case: ordinary and relative logarithmic differentials agree when the value groups agree. We prove this below. For conductors we apply the existing prime-degree theory to character quotients and recover the upper filtration when G is elementary abelian. This reconstruction does not require purity. In a general finite group, order- p characters do not suffice to separate subgroups; already for $G \simeq \mathbb{Z}/p^2\mathbb{Z}$, all such characters vanish on pG . Higher-order characters require additional parameter and conductor theory. Thus the limitation here concerns the formulas obtained from purity, rather than the existence of a general upper ramification theory over henselian valuation rings [14].

6.1. Prime-degree parameter ideals. The parameter descriptions in this section are statements over henselian valuation rings. The equal-characteristic norm and refined-conductor theorems of [12] assume either rank one or defectlessness. The mixed-characteristic results of [13] allow arbitrary rank, and Section 8 of that paper extends the method to equal characteristic.

For henselian cyclic degree- p extensions, [12, Theorem 0.3] and [13, Theorem 1.3] identify the Swan ideal $\mathcal{H} \subseteq \mathcal{O}_K$ with

$$(10) \quad \mathcal{H} = (N_{L/K} \mathcal{J}_\sigma) \mathcal{O}_K.$$

In equal characteristic, $\mathcal{H}$ is generated by the reciprocals of nonzero Artin–Schreier parameters defining the extension. In the Kummer case, with $z = \zeta_p - 1$, it is generated by $z^p/(h-1)$ for nonzero parameters h defining the extension, with $h \neq 1$. Thus Proposition 4.6 recovers the value-set part of these theorems in the pure degree- p cases:

$$v\mathcal{H} = \text{Final}_{vK}(p(\delta - D_1(\theta, K))).$$

Formula (7) extends this norm calculation to arbitrary pure Galois degree and to each subgroup ideal. It does not supply an Artin–Schreier or Kummer parameter description of those higher-degree norm ideals, nor a refined Swan conductor for them.

Proposition 3.10 and Corollary 4.5 of [12], and Proposition 3.8 and Corollary 4.4 of [13], relate defectlessness to principality of the logarithmic ideal and of the Swan ideal (with the stated hypotheses). For a pure immediate unbranched extension of arbitrary degree, $D_1(\theta, K)$ has no maximum, so every nontrivial $I_H = I_{\beta_H - D_1(\theta, K)}$ is nonprincipal. The same implication is not asserted for general higher-degree Galois extensions without purity.

6.2. Logarithmic differentials and the upper filtration.

Proposition 6.1. *Let $(L/K, v)$ be an extension of valued fields with $vL = vK$. The natural map*

$$\Omega_{\mathcal{O}_L/\mathcal{O}_K} \longrightarrow \Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log}$$

is an isomorphism. In particular, for a finite separable, immediate, unbranched extension pure on θ ,

$$\Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log} \simeq I_\alpha/I_\beta, \quad \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log} = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K}.$$

Proof. For $x \in L^\times$, choose $a \in K^\times$ with $va = vx$ and write $x = au$, where $u \in \mathcal{O}_L^\times$. Define

$$\ell(x) = u^{-1}du \in \Omega_{\mathcal{O}_L/\mathcal{O}_K}.$$

This does not depend on the choice of a : replacing a by another such element multiplies u by a unit of $\mathcal{O}_K$, whose differential vanishes. The Leibniz rule gives $\ell(xy) = \ell(x) + \ell(y)$, and $\ell(a) = 0$ for $a \in K^\times$. If $x \in \mathcal{O}_L \setminus \{0\}$, the chosen a lies in $\mathcal{O}_K$, so $dx = a du = x\ell(x)$. These are precisely the additional relations for relative logarithmic differentials. Consequently $d \log x \mapsto \ell(x)$, together with $dx \mapsto dx$, defines an inverse

to the natural map: in the logarithmic module, $d \log x = d \log u = u^{-1} du$. The final assertion follows from (1). $\square$

The proposition uses equality of value groups, not purity. If the value group increases, an arbitrary $x \in L^\times$ need not be a K -multiple of an integral unit, so this proof does not apply. The ordinary differential quotient alone therefore does not provide the same identification for every pure extension. Nor does the identification above construct a refined Swan map, whose definition also involves a character and its parameter ideal.

Theorems 0.5 and 1.5 of [12] and [13] construct a refined Swan conductor and an isomorphism

$$\Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log} / \mathcal{J}_\sigma \Omega_{\mathcal{O}_L/\mathcal{O}_K}^{\log} \simeq \mathcal{J}_\sigma / \mathcal{J}_\sigma^2,$$

compatible with norm maps. These are cyclic degree- p statements under the hypotheses specified above. Proposition 6.1 identifies the differential modules when $vL = vK$, but does not extend these particular character and norm maps to arbitrary degree.

[14, Theorem 1.1] characterizes the Swan ideal by the logarithmic upper ramification filtration. If $\overline{\mathcal{O}}$ is the valuation ring in a separable closure and $\mathfrak{a}$ is a nonzero proper ideal of it, the image of $G_{\log}^{\mathfrak{a}}$ in a cyclic degree- p quotient is the whole quotient exactly when

$$\mathfrak{a} \cap \mathcal{O}_K \supseteq \mathcal{H};$$

it is trivial otherwise. Combined with the prime-degree norm identity of [12, 13], this relates the norm of the prime-degree logarithmic ramification ideal to an upper-filtration threshold. The upper filtration and the subgroup ideals I_H are distinct constructions. Section 7.2.4 applies this criterion to the depth-one defect example.

Finally, Section 1.5 and the example in Section 10.4 of [14] show why all ideals, rather than just principal ideals, are needed as indices: the cuts represented by $a\overline{\mathcal{O}}$ and $a\mathfrak{m}_{\overline{\mathcal{O}}}$ can give different thresholds although principal test ideals do not distinguish them. Thus the full ideal index set retains endpoint information that is absent from the positive principal indices.

6.3. Reconstruction from character quotients. Let M/F be an elementary abelian p -extension of henselian valued fields with residue characteristic p . Put $A = \mathcal{O}_F$ and $G = \text{Gal}(M/F)$. For a nonzero character $\chi : G \rightarrow \mathbb{F}_p$, let $\mathcal{H}_\chi$ be the Swan ideal of $M^{\ker \chi}/F$. Let $\overline{A}$ be the valuation ring in a separable closure, and let $U_{\mathfrak{a}}$ denote the image in G of the logarithmic upper group indexed by a nonzero proper ideal $\mathfrak{a}$ of $\overline{A}$.

Proposition 6.2. *With the above notation,*

$$(11) \quad U_{\mathfrak{a}} = \bigcap_{\substack{\chi \neq 0 \\ \mathfrak{a} \cap A \subsetneq \mathcal{H}_{\chi}}} \ker \chi,$$

where an empty intersection denotes G .

Proof. By [14, Theorem 1.1], the image of $U_{\mathfrak{a}}$ in $G/\ker \chi$ is trivial exactly when $\mathfrak{a} \cap A \subsetneq \mathcal{H}_{\chi}$. A subgroup of an elementary abelian group is the intersection of the kernels of all characters vanishing on it. Applying this elementary duality proves the formula. $\square$

7. PURE EXTENSIONS OF DEGREE p^2 : TWO EXAMPLES

7.1. A totally ramified Artin–Schreier compositum.

Example 7.1. Let k be a perfect field of characteristic $p > 0$, let $K = k((t))$ with $vt = 1$, and put $n = p^2$. Consider

$$\begin{aligned} u^p - u &= t^{-1}, \\ w^p - w &= t^{-(p+1)}, \\ L &= K(u, w). \end{aligned}$$

Recall that replacing a root y of $y^p - y = f$ by $y - a$, with $a \in K$, replaces the parameter by $f - (a^p - a)$. An element $a^p - a$ is called an Artin–Schreier coboundary, and the parameter classes lie in the $\mathbb{F}_p$ -vector space $K/\wp(K)$, where $\wp(K) = \{a^p - a : a \in K\}$. If $v(a^p - a) < 0$, then $va < 0$ and

$$v(a^p - a) = pva,$$

since $pva < va$. Its pole order is therefore divisible by p . On the other hand, for $(A, B) \in \mathbb{F}_p^2 \setminus \{(0, 0)\}$,

$$v\left(At^{-1} + Bt^{-(p+1)}\right) = \begin{cases} -1, & B = 0, \\ -(p+1), & B \neq 0. \end{cases}$$

Neither pole order is divisible by p . Hence no nonzero combination of the two parameters lies in $\wp(K)$, proving that their classes are linearly independent. Artin–Schreier theory now gives $[L : K] = n$ and

$$G = \{\rho_{i,j} : u \mapsto u + i, w \mapsto w + j \mid i, j \in \mathbb{F}_p\} \simeq C_p \times C_p.$$

This is a degree- p^2 Artin–Schreier compositum; a single Artin–Schreier equation would have degree at most p .

An explicit pure generator is obtained by setting

$$z = w - u^{p+1}, \quad \theta = (tz)^{-1}.$$

Indeed,

$$z^p - z = -ut^{-p} - ut^{-1} - t^{-2}.$$

Since $vu = -1/p$, the first term has strictly smaller value than the others. Therefore

$$vz = -1 - \frac{1}{n}, \quad v\theta = \frac{1}{n}.$$

It follows that $e(L/K) = n$, so L/K is totally ramified, $L = K(\theta)$, and θ is a uniformizer of $\mathcal{O}_L$. Its minimal polynomial g is Eisenstein and $\mathcal{O}_L = \mathcal{O}_K[\theta]$. Moreover, for $f = \sum_{j=0}^{n-1} a_j X^j$, the nonzero terms $a_j \theta^j$ have distinct values modulo vK . Thus $v(f(\theta)) = \min_j (va_j + j/n)$, which proves purity on θ . The approximation set $D_1(\theta, K)$ has maximum $1/n$.

The conjugate distances can be computed directly. We have

$$\rho_{i,j}z - z = j - 2iu - it^{-1} - i^2,$$

whose value is -1 if $i \neq 0$, and zero if $i = 0, j \neq 0$. This remains valid for $p = 2$, when the term $2iu$ vanishes. Since every conjugate of z has value $-1 - 1/n$,

$$\beta_{i,j} = v(\rho_{i,j}\theta - \theta) = \begin{cases} 2/n, & i \neq 0, \\ 1 + 2/n, & i = 0, j \neq 0. \end{cases}$$

There are $n - p$ conjugates of the first type and $p - 1$ of the second. Consequently, for $H = \{\rho_{0,j} : j \in \mathbb{F}_p\}$,

$$\begin{aligned} I_G &= \mathfrak{m}_L = (\theta)\mathcal{O}_L, \\ I_H &= t\mathfrak{m}_L, \\ I_J &= \mathfrak{m}_L \quad (J \neq \{1\}, J \not\subseteq H). \end{aligned}$$

The minimum distance determines I_G , while the derivative retains both distance levels:

$$v(g'(\theta)) = (n - p)\frac{2}{n} + (p - 1)\left(1 + \frac{2}{n}\right) = p + 1 - \frac{2}{n}.$$

Writing $d = (p + 1)n - 2$, monogenicity gives

$$\begin{aligned} \mathfrak{D}_{\text{prim}} &= \mathfrak{D} = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} = \mathfrak{m}_L^d, \\ \Omega_{\mathcal{O}_L/\mathcal{O}_K} &\simeq \mathcal{O}_L/\mathfrak{m}_L^d. \end{aligned}$$

Since $d > n - 1$, we obtain

$$\mathfrak{D}_{\text{prim}} = \mathfrak{D} = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} \subsetneq I_G^{n-1}.$$

The classical trace formula, with $e = n$ and different exponent d , gives

$$\begin{aligned}\mathrm{Tr}_{L/K}(\mathcal{O}_L) &= \mathfrak{m}_K^{\lfloor d/n \rfloor} = t^p \mathcal{O}_K, \\ T(\mathcal{O}_L/\mathcal{O}_K) &= t^p \mathcal{O}_L \subsetneq I_G^{n-1}.\end{aligned}$$

The norm ideals are

$$(\mathrm{N}_{L/K} I_G) \mathcal{O}_K = t \mathcal{O}_K, \quad (\mathrm{N}_{L/K} I_H) \mathcal{O}_K = t^{n+1} \mathcal{O}_K.$$

Finally, the lower ramification breaks are 1 and $n + 1$, since for a uniformizer the lower index is $n\beta_{i,j} - 1$. Herbrand's change of numbering gives upper breaks 1 and $p + 1$. Equivalently, the character $\chi_{a,b}(\rho_{i,j}) = ai + bj$ is represented by $at^{-1} + bt^{-(p+1)}$, and its Swan conductor is 1 when $b = 0$, $a \neq 0$, and $p + 1$ when $b \neq 0$. These are the classical Artin–Schreier and upper-numbering computations of [1].

This example disproves, for pure extensions of general degree, each naive replacement of $p - 1$ by $n - 1$ in

$$\begin{aligned}T(\mathcal{O}_L/\mathcal{O}_K) &= I_G^{p-1}, \\ \mathfrak{D}_{\mathrm{prim}} &= I_G^{p-1}, \\ \mathfrak{D} &= I_G^{p-1}, \\ \mathrm{ann} \Omega_{\mathcal{O}_L/\mathcal{O}_K} &= I_G^{p-1}.\end{aligned}$$

In this example $\mathfrak{D} = \mathrm{ann} \Omega$, although neither ideal equals the proposed power of the largest subgroup ramification ideal.

7.2. An immediate elementary abelian extension. To examine purity in the presence of defect beyond prime degree, we use the depth-one Hahn-field construction of [15, Section 5.2]. Let $k = \overline{\mathbb{F}_p}$ and choose $r_1 = p < r_2 < \cdots$ tending to infinity such that $\{\pi\} \cup \{1/r_i : i \geq 1\}$ is linearly independent over $\mathbb{Q}$, where π is a fixed positive irrational real number. In $k((t^{\mathbb{R}}))$, put

$$\begin{aligned}a_\ell &= t^{-\pi/p^\ell}, \\ b_\ell &= t^{-1/r_\ell} - t^{-1/r_{\ell+1}}, \\ K &= k(t, t^\pi)(a_\ell, b_\ell : \ell \geq 1).\end{aligned}$$

Write $c_\ell = \sum_{j=1}^\ell a_j$ and $d_\ell = \sum_{j=1}^\ell b_j$.

Let u and w be the negative-support Hahn representatives of the Artin–Schreier roots below. With $vt = 1$, put

$$\begin{aligned} u^p - u &= t^{-\pi}, \\ w^p - w &= t^{-p-1}, \\ \theta &= u + tw, \\ L &= K(\theta) = K(u, w). \end{aligned}$$

Thus K has *no finite limits*: the support of each of its elements meets every interval $(-\infty, M)$ in a finite set. The two degree- p extensions are linearly disjoint, and the approximation calculations of [15, Section 5.2] give

$$\begin{aligned} d_1(u, K) &= 0^-, \\ d_1(w, K) &= -1^-, \\ d_1(\theta, K) &= 0^-, \\ \text{dep}(\theta) &= 1. \end{aligned}$$

In particular, L/K is pure on θ and has degree p^2 . The approximation cut is balanced because $d_1(u, K) = vt + d_1(w, K)$. For the finite approximants c_ℓ and d_ℓ of that construction,

$$v(\theta - (c_\ell + d_\ell)) = \min \left\{ -\frac{\pi}{p^{\ell+1}}, -\frac{1}{r_{\ell+1}} \right\} \longrightarrow 0^-.$$

No approximation from K has nonnegative value, since the negative support of θ is infinite. The group vK is dense in $\mathbb{R}$, since it contains $\mathbb{Z} + \pi\mathbb{Z}$.

The field K is not assumed henselian. We verify the valuation hypotheses before using the general formulas.

Lemma 7.2. *The chosen Hahn valuation makes L/K immediate and unibranched.*

Proof. For $z = P(\theta) \neq 0$ of degree less than p^2 , choose $c \in K$ with $v_c(P) = vz$. Choose a better approximation $d \in K$, so that $\delta_d = v(\theta - d) > \delta_c = v(\theta - c)$. In the expansion $P(X) = \sum_j P_{j,c}(X - c)^j$, the difference of powers satisfies

$$v((\theta - c)^j - (d - c)^j) \geq (j - 1)\delta_c + \delta_d > j\delta_c \quad (j \geq 1).$$

Hence $v(z - P(d)) > vz$. Every nonzero element of L consequently has an approximation in K of the same value and the same leading residue. This proves $vL = vK$ and $Lv = Kv$.

For $\sigma \in G$, we have $v(\sigma\theta - \theta) \geq 0$, whereas $v(\theta - c) < 0$ for every $c \in K$. Therefore $v(\sigma\theta - c) = v(\theta - c)$, and the same expansion gives

$v(\sigma z) \geq v_c(P) = vz$. Iterating σ , which has finite order, forces equality. The valuation is thus G -invariant. Since the Galois group acts transitively on the prolongations of a valuation to a finite normal extension, there is only one such prolongation. This proves unibranchedness. $\square$

Write $\Gamma = vK = vL$. The invariant-valuation formulation of Theorem 3.1 therefore applies directly.

Theorem 7.3. *For the extension just constructed, $[L : K] = p^2$, $vL = vK$, $Lv = Kv$, and θ has depth one. The chosen valuation is unibranched and the defect is p^2 . Write $H_1 = \{\rho_{0,j} : j \in \mathbb{F}_p\}$, where $\rho_{i,j}(\theta) = \theta + i + tj$. Then*

$$\begin{aligned} \Omega_{\mathcal{O}_L/\mathcal{O}_K} &\simeq \mathfrak{m}_L/t^{p-1}\mathfrak{m}_L, \\ \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K} &= t^{p-1}\mathcal{O}_L, \\ I_{H_1} &= t\mathfrak{m}_L, \\ I_H &= \mathfrak{m}_L \quad (\{1\} \neq H \neq H_1), \\ \mathfrak{D}_{\text{prim}} = T &= t^{p-1}\mathfrak{m}_L, \\ \text{Tr}_{L/K}(\mathcal{O}_L) &= t^{p-1}\mathfrak{m}_K, \\ \mathfrak{C} &= t^{1-p}\mathcal{O}_L, \\ \mathfrak{D} &= t^{p-1}\mathcal{O}_L. \end{aligned}$$

Thus the primitive-generator different and trace ideal have open value cuts, whereas the Dedekind different and differential annihilator have closed cuts.

Let $F = K^h$, $E = LF$, and $A = \mathcal{O}_F$. The degree and defect remain p^2 . The character $\chi_{a,b}(\rho_{i,j}) = ai + bj$ has Swan ideal

$$\mathcal{H}_{a,b} = \begin{cases} \mathfrak{m}_A, & b = 0, a \neq 0, \\ t^p\mathfrak{m}_A, & b \neq 0. \end{cases}$$

For a nonzero proper ideal $\mathfrak{a}$ in the valuation ring of a separable closure of F , the logarithmic upper image is G if $\mathfrak{a} \cap A \supseteq \mathfrak{m}_A$, is H_1 if $t^p\mathfrak{m}_A \subseteq \mathfrak{a} \cap A \subsetneq \mathfrak{m}_A$, and is trivial otherwise. In contrast, the full-extension norm ideals of I_G and I_{H_1} are respectively $\mathfrak{m}_A$ and $t^{p^2}\mathfrak{m}_A$.

The differential, trace, different, and subgroup-ideal computations are proved in the following subsections. Proposition 7.4 also verifies degree preservation under henselization and computes the character ideals; Section 7.2.5 proves the filtration and norm assertions. These computations together prove Theorem 7.3.

7.2.1. *The minimal polynomial and differential module.* Let $f = t^{-\pi}$, $h = t^{-p-1}$, and $q = t^p - t$. Since

$$\theta^p - \theta - f = qw + t^p h,$$

the monic minimal polynomial of θ is

$$\begin{aligned} Y(X) &= X^p - X - f - t^p h, \\ g(X) &= Y(X)^p - q^{p-1} Y(X) - q^p h. \end{aligned}$$

It has degree p^2 and vanishes at θ , so it is indeed the minimal polynomial. Its derivative is constant:

$$\begin{aligned} g'(X) &= q^{p-1}, \\ vq &= 1, \\ v(g'(\theta)) &= p - 1. \end{aligned}$$

Equivalently, the conjugates are $\theta + i + tj$, with $i, j \in \mathbb{F}_p$, and

$$v(i + tj) = \begin{cases} 0, & i \neq 0, \\ 1, & i = 0, j \neq 0. \end{cases}$$

There are $p(p-1)$ differences of value zero and $p-1$ of value one, which again gives the derivative sum $p-1$.

For $a \in K$ put $d = v(\theta - a) < 0$ and $Q = X - a$. Every root $\theta + i + tj$ has distance d from a . Factoring g shows $v(g(a)) = p^2 d$. The coefficient of Q^k has value at least $(p^2 - k)d$, since it is a sum of products of $p^2 - k$ such root differences. Consequently every term of the Q -expansion has truncated value at least $p^2 d$, with equality for the constant and leading terms. Hence

$$\begin{aligned} \nu_Q(g) &= p^2 d, \\ \alpha_Q &= -d, \\ \beta_Q &= (p-1) - p^2 d. \end{aligned}$$

Taking the final segments generated by a cofinal linear family yields

$$(12) \quad \begin{aligned} \alpha &= \Gamma_{>0}, & I_\alpha &= \mathfrak{m}_L, \\ \beta &= \Gamma_{>p-1}, & I_\beta &= t^{p-1} \mathfrak{m}_L. \end{aligned}$$

Therefore the Kähler module and its annihilator are

$$(13) \quad \begin{aligned} \Omega_{\mathcal{O}_L/\mathcal{O}_K} &\simeq \mathfrak{m}_L/t^{p-1} \mathfrak{m}_L \neq 0, \\ \text{ann}_{\mathcal{O}_L} \Omega_{\mathcal{O}_L/\mathcal{O}_K} &= t^{p-1} \mathcal{O}_L. \end{aligned}$$

Indeed, $\gamma + \Gamma_{>0} \subseteq \Gamma_{>p-1}$ holds exactly when $\gamma \geq p-1$. The strict cuts defining the numerator and denominator therefore give a closed cut for the annihilator. Nonzero quotient representatives have values $0 < vx \leq p-1$.

The differential module and its annihilator

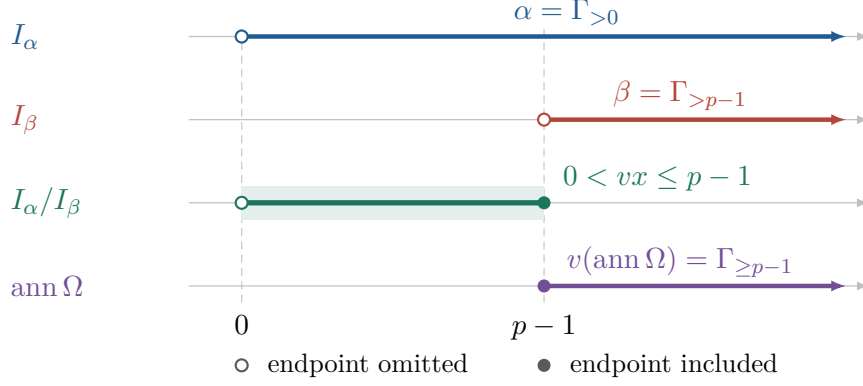


FIGURE 5. The immediate elementary abelian example has $\Omega \simeq \mathfrak{m}_L/t^{p-1}\mathfrak{m}_L$. The denominator omits its lower endpoint, so a quotient representative can have value $p-1$. The annihilator $t^{p-1}\mathcal{O}_L$ includes that endpoint. All four rows use the same value coordinates.

7.2.2. *Trace ideals and the two differents.* The derivative $g' = q^{p-1}$ is constant of value $p-1$. The approximation values $\delta_c = v(\theta - c)$ are cofinal below zero. Theorem 4.1 therefore gives

$$\mathfrak{D}_{\text{prim}} = t^{p-1}\mathfrak{m}_L.$$

For completeness, the trace calculation is equally explicit. Writing $n = p^2$ and $u_c = t_c(\theta - c)$, the only nonzero power trace among $0 \leq i < n$ is

$$\text{Tr}(u_c^{n-1}) = t_c^{n-1}q^{p-1}, \quad v \text{Tr}(u_c^{n-1}) = (p-1) - (n-1)\delta_c.$$

These values are coinitial above $p-1$, and hence

$$\text{Tr}_{L/K}(\mathcal{O}_L) = t^{p-1}\mathfrak{m}_K, \quad T = t^{p-1}\mathfrak{m}_L = \mathfrak{D}_{\text{prim}}.$$

Theorem 5.1 yields

$$\begin{aligned} \mathfrak{C} &= \mathcal{O}_L :_L (t^{p-1}\mathfrak{m}_L) = t^{1-p}\mathcal{O}_L, \\ \mathfrak{D} &= \mathfrak{D}_{\text{prim}}^v = t^{p-1}\mathcal{O}_L. \end{aligned}$$

Together with (13), this proves

$$T = \mathfrak{D}_{\text{prim}} \subsetneq \mathfrak{D} = \text{ann } \Omega_{\mathcal{O}_L/\mathcal{O}_K}.$$

The inclusion is strict because the endpoint is unattained in $v\mathfrak{D}_{\text{prim}} = \Gamma_{>p-1}$.

The all-elements convention has a different answer here. The subextension $K(u)/K$ has approximation cut 0^- ; its integral normalized

generators have derivative values cointial above zero. Their derivatives therefore generate $\mathfrak{m}_L$ after extension to $\mathcal{O}_L$. Conversely, every integral $b \in L \setminus K$ and every distinct K -conjugate b' have the same residue: the Galois action preserves the valuation and acts trivially on $Lv = Kv$. Thus $v(b - b') > 0$, and the product formula for the minimal-polynomial derivative gives $v(h'_b(b)) > 0$. Consequently

$$\mathfrak{D}_{\text{all}} = \mathfrak{m}_L \supsetneq \mathfrak{D}_{\text{prim}} = t^{p-1}\mathfrak{m}_L.$$

Hence the two derivative-ideal conventions differ in composite degree.

7.2.3. Subgroup ramification ideals. Write $G = \text{Gal}(L/K) \simeq \mathbb{F}_p^2$ and let $\rho_{i,j}$ send u to $u + i$ and w to $w + j$. Then $\rho_{i,j}(\theta) - \theta = i + tj$. The subgroup

$$H_1 = \{\rho_{0,j} : j \in \mathbb{F}_p\} = \text{Gal}(L/K(u))$$

is the unique order- p subgroup whose nonidentity elements all have conjugate-distance value one. Every other nontrivial subgroup contains an element with $i \neq 0$. Thus

$$\beta_H = \begin{cases} 1, & H = H_1, \\ 0, & H \neq H_1, H \neq \{1\}. \end{cases}$$

Since $D_1(\theta, K)$ has cut 0^- , the pure ramification formula gives

$$(14) \quad \begin{aligned} I_{H_1} &= I_{\Gamma_{>1}} = t\mathfrak{m}_L, \\ I_H &= I_{\Gamma_{>0}} = \mathfrak{m}_L \quad (H \neq H_1, H \neq \{1\}). \end{aligned}$$

Let $\text{Ram}(L/K, v) = \{I_H : \{1\} \neq H \leq G\}$. Then

$$\begin{aligned} \text{Ram}(L/K, v) &= \{t\mathfrak{m}_L, \mathfrak{m}_L\}, \\ t\mathfrak{m}_L &\subsetneq \mathfrak{m}_L, \# \text{Ram}(L/K, v) = 2 > 1 = \text{dep}(\theta). \end{aligned}$$

Both ideals are nonprincipal because their value segments have no least element. The ramification calculation separates the two conjugate levels, whereas the differential denominator combines their multiplicities through $v(g'(\theta)) = p - 1$. In this example one also has $I_\beta = I_{H_1}^{p-1} = t^{p-1}\mathfrak{m}_L$, since $\mathfrak{m}_L^{p-1} = \mathfrak{m}_L$ in the dense value group.

7.2.4. Henselization and character Swan ideals. Let K, u, w , and $L = K(u, w)$ be as in Section 7.2, and put

$$\begin{aligned} F &= K^h, \\ E &= LF, \\ A &= \mathcal{O}_F. \end{aligned}$$

The henselization is taken with respect to the chosen Hahn valuation. A rank-one valued field is dense in its henselization [16]. If z has

Two subgroup ramification ideals

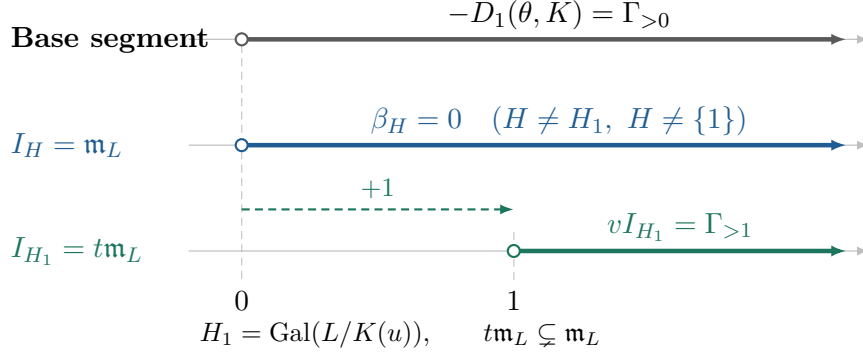


FIGURE 6. The subgroup minima zero and one give the open rays $v\mathfrak{m}_L = \Gamma_{>0}$ and $v(tm_L) = \Gamma_{>1}$. The subgroup H_1 gives the translated segment $\Gamma_{>1}$.

bounded approximation values over K , it therefore lies outside F and has the same approximation cut over F : for $a \in F$, choose $a_0 \in K$ with $v(a - a_0) > v(z - a)$, giving $v(z - a_0) = v(z - a)$.

Proposition 7.4. *The extension E/F is immediate Galois of degree p^2 . Its $p + 1$ degree- p subextensions are*

$$E_\infty = F(u), \quad E_\lambda = F(w + \lambda u) \quad (\lambda \in \mathbb{F}_p).$$

They have Artin–Schreier parameters

$$f_\infty = t^{-\pi}, \quad f_\lambda = t^{-p-1} + \lambda t^{-\pi},$$

approximation cuts 0^- and -1^- , respectively, and Swan ideals

$$(15) \quad \mathcal{H}_\infty = \mathfrak{m}_A, \quad \mathcal{H}_\lambda = t^p \mathfrak{m}_A \quad (\lambda \in \mathbb{F}_p).$$

All these subextensions have defect p . Both Swan value segments have unattained lower endpoints.

Proof. For $\lambda \neq 0$, choose $c \in K$ with $v(u - c) > -1$. An approximation to $w + \lambda u$ of value at least -1 would, after subtracting λc , approximate w to value at least -1 , contradicting $d_1(w, K) = -1^-$. Conversely, put $z_{\lambda, \ell} = t^{-1}d_\ell + \lambda c_\ell$. For sufficiently large ℓ ,

$$v(w + \lambda u - z_{\lambda, \ell}) = -1 - \frac{1}{r_{\ell+1}} \longrightarrow -1^-.$$

For $\lambda = 0$ this is the calculation for w . Density preserves these cuts over F , as well as the cut 0^- for u . Every nonzero $\mathbb{F}_p$ -linear combination of u and w has bounded approximation values, so no such combination belongs to F . Their Artin–Schreier classes remain independent over

F . It follows that $[E : F] = p^2$ and that the stated list of subextensions is complete. The decomposition group of the chosen valuation on L/K is consequently all of G ; hence $E = LK^h = L^h$. Since L/K and henselizations are immediate, E/F is immediate. Every degree- p intermediate extension therefore has defect p .

For $(a, b) \in \mathbb{F}_p^2 \setminus \{(0, 0)\}$, write

$$\begin{aligned} z_{a,b} &= au + bw, \\ f_{a,b} &= at^{-\pi} + bt^{-p-1}, \\ \rho_{a,b} &= \sup v(z_{a,b} - F). \end{aligned}$$

Then $\rho_{a,b} = 0$ if $b = 0$ and $\rho_{a,b} = -1$ otherwise. The character $\chi_{a,b}(\rho_{i,j}) = ai + bj$ is represented by the Artin–Schreier class of $f_{a,b}$. The parameters representing this fixed character are $f_{a,b} - (c^p - c)$, $c \in F$, and satisfy

$$v(f_{a,b} - (c^p - c)) = p v(z_{a,b} - c),$$

since the latter approximation values are negative. The parameter description of the Swan ideal gives

$$(16) \quad \mathcal{H}_{a,b} = \left(\frac{1}{f_{a,b} - (c^p - c)} : c \in F \right) A = \{0\} \cup \{x \in F^\times : vx > -p\rho_{a,b}\}.$$

Nonzero scalar multiples of a character give the same ideal. This proves (15). For explicit cofinal parameter families, take $c = c_\ell$ on the independent line and $c = z_{\lambda,\ell}$ on the other lines. The reciprocal parameter values are

$$\frac{\pi}{p^\ell} \longrightarrow 0^+, \quad p + \frac{p}{r_{\ell+1}} \longrightarrow p^+,$$

respectively. Neither limiting endpoint is attained. $\square$

The prime-degree logarithmic ideals and ordinary differential modules are therefore

$$\begin{aligned} \mathcal{J}_\infty &= \mathfrak{m}_{E_\infty}, \\ \Omega_{\mathcal{O}_{E_\infty}/A} &= 0, \\ \mathcal{J}_\lambda &= t\mathfrak{m}_{E_\lambda}, \\ \Omega_{\mathcal{O}_{E_\lambda}/A} &\simeq t\mathfrak{m}_{E_\lambda}/t^p\mathfrak{m}_{E_\lambda}. \end{aligned}$$

Their norms to F are the ideals of (15), in agreement with the prime-degree norm identity of [12, 13]. The differential annihilators are $\mathcal{O}_{E_\infty}$ and $t^{p-1}\mathcal{O}_{E_\lambda}$. The independent line has Swan cut of infimum zero, whereas the p dependent lines have Swan cut of infimum p . These numerical infima do not encode the unattained endpoints of the conductor ideals.

The refined Swan maps of [12, Theorem 0.5] send

$$\left[\frac{1}{f_{a,b} - (c^p - c)} \right] \mapsto \left[d \log(f_{a,b} - (c^p - c)) \right]$$

from $\mathcal{H}_{a,b}/\mathcal{H}_{a,b}^2$ to $\Omega_A^{\log}/(\mathcal{I}_{a,b} \cap A)\Omega_A^{\log}$, where $\mathcal{I}_{a,b}$ is the ordinary difference ideal of the corresponding degree- p subextension. For $b = 0$ the source is zero, since $\mathfrak{m}_A^2 = \mathfrak{m}_A$. For $b \neq 0$ it is $t^p \mathfrak{m}_A/t^{2p} \mathfrak{m}_A$. These formulas determine the sources of the refined maps. Their images are not computed here.

7.2.5. *The logarithmic upper filtration and norm ideals.* For the extension E/F of Proposition 7.4, put $H_1 = \ker \chi_{1,0} = \{\rho_{0,j} : j \in \mathbb{F}_p\}$. Equations (15) and (11) give

$$(17) \quad U_{\mathfrak{a}} = \begin{cases} G, & \mathfrak{a} \cap A \supseteq \mathfrak{m}_A, \\ H_1, & t^p \mathfrak{m}_A \subseteq \mathfrak{a} \cap A \subsetneq \mathfrak{m}_A, \\ \{1\}, & \mathfrak{a} \cap A \subsetneq t^p \mathfrak{m}_A. \end{cases}$$

For principal indices $\mathfrak{a}_r = \{x \in \overline{A} : vx \geq r\}$, $r \in v(\overline{F}^\times)$, $r > 0$, the image is H_1 for $0 < r \leq p$ and is trivial for $r > p$. The full group occurs at the nonprincipal index $\mathfrak{m}_{\overline{A}}$. At $r = p$ the closed index contains the Swan ideal with the open cut $v > p$.

The subgroup norm ideals of the full extension have different thresholds. Their value segments are preserved by henselization: L is dense in $E = L^h$, and a sufficiently close $y \in L$ to $x \in E^\times$, with $\sigma x \neq x$, satisfies $v((\sigma y - y)/y) = v((\sigma x - x)/x)$. Hence (14) and Proposition 4.6 give

$$(N_{E/F} I_G)A = \mathfrak{m}_A, \quad (N_{E/F} I_{H_1})A = t^{p^2} \mathfrak{m}_A.$$

The positive character Swan ideals instead equal $t^p \mathfrak{m}_A$, because their norm maps originate in the degree- p quotient fields. In particular, the norm of I_G alone does not determine the positive upper threshold.

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